Evaluation of increment PDFs using Laplace's method

1 Starting point: the distribution of h

We start from Equation (11) of [1], which defines the probability density function of the local Hölder exponent h:

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)}}{\displaystyle\int_{h_{\min}}^{h_{\max}} \left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)} \, \mathrm{d}h}. \tag{1}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h-c_1)^2}{2c_2},$$

with $h_{\min} = -\infty$ and $h_{\max} = +\infty$, we obtain

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{(h-c_1)^2}{2c_2} \ln\left(\frac{L}{\ell}\right)\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{(h-c_1)^2}{2c_2} \ln\left(\frac{L}{\ell}\right)\right] \mathrm{d}h}. \tag{2}$$

To simplify notation, we define

$$\alpha = -\ln\left(\frac{\ell}{L}\right) > 0. \tag{3}$$

Then the expression becomes

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right] \mathrm{d}h}. \tag{4}$$

The denominator is a Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{\alpha}{2c_2}(h-c_1)^2} \mathrm{d}h = \sqrt{\frac{2\pi c_2}{\alpha}}. \tag{5}$$

Hence,

$$\boxed{\mathcal{P}_h^{(\ell)}(h) = \sqrt{\frac{\alpha}{2\pi c_2}} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right].} \tag{6}$$

2 Computation of the PDF using Laplace's method

2.1 General form of the PDF

We now consider the probability density function of velocity increments:

$$\mathcal{P}_{\delta_\ell u}(\delta_\ell u) = \frac{1}{2\pi\sigma} \sqrt{\frac{\alpha}{c_2}} \int_{-\infty}^{+\infty} \exp[-f(h, \alpha, \delta_\ell u)] \, \mathrm{d}h. \tag{7}$$

with

$$f(h, \alpha, \delta_\ell u) = \alpha\left(\frac{(h-c_1)^2}{2c_2} - h\right) + \frac{1}{2}\left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h}. \tag{8}$$

Laplace approximation and signs. The integral is dominated by the minimum of $f(h, \alpha, \delta_\ell u)$. A quadratic expansion around the stationary point $h^\star$ gives

$$\int_{-\infty}^{+\infty} \exp[-f(h, \alpha, \delta_\ell u)]\, \mathrm{d}h \approx \exp[-f(h^\star, \alpha, \delta_\ell u)] \sqrt{\frac{2\pi}{f''(h^\star, \alpha, \delta_\ell u)}}. \tag{9}$$

Therefore

$$\boxed{\mathcal{P}_{\delta_\ell u}(\delta_\ell u) \approx \frac{1}{2\pi\sigma} \sqrt{\frac{\alpha}{c_2}} \sqrt{\frac{2\pi}{f''(h^\star, \alpha, \delta_\ell u)}}\, \exp[-f(h^\star, \alpha, \delta_\ell u)].} \tag{10}$$

Since $\alpha > 0$, the term $\alpha(h - c_1)^2/(2c_2)$ grows quadratically for large $|h|$, and the exponential term is positive. Thus $f(h, \alpha, \delta_\ell u) \to +\infty$ as $|h| \to \infty$, so the stationary point is a *minimum* and $f''(h^\star, \alpha, \delta_\ell u) > 0$, which justifies the steepest descent method.

Knowing that

$$f''(h, \alpha, \delta_\ell u) = \frac{\partial^2 f}{\partial h^2}(h, \alpha, \delta_\ell u) = \frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h}, \tag{11}$$

we finally obtain

$$\boxed{\begin{aligned} \mathcal{P}_{\delta_\ell u}(\delta_\ell u) \approx{}& \frac{1}{2\pi\sigma} \sqrt{\frac{\alpha}{c_2}} \sqrt{\frac{2\pi}{\frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h^\star}}} \\ &\times \exp\left[-\alpha\left(\frac{(h^\star - c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}\left(\frac{\delta_\ell u}{\sigma}\right)^2 e^{2\alpha h^\star}\right]. \end{aligned}} \tag{12}$$

2.2 Normalized PDF

The p-th order moment of the velocity increment is expressed as

$$\langle |\delta_\ell u|^p \rangle \sim \ell^{\zeta_p}, \quad \text{with} \quad \zeta_p = \min_h \big[ph + 1 - \mathcal{D}^E(h)\big]. \tag{13}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h - c_1)^2}{2c_2}, \tag{14}$$

the minimization of Eq. (21) gives

$$\frac{\mathrm{d}}{\mathrm{d}h}\big[ph + 1 - \mathcal{D}^E(h)\big] = 0 \quad \Longrightarrow \quad p = \frac{h - c_1}{c_2}. \tag{15}$$

Hence, the minimizing exponent is

$$h_p = c_1 + pc_2, \tag{16}$$

and inserting this value into Eq. (21) gives

$$\zeta_p = pc_1 + c_2 p^2. \tag{17}$$

In particular, for the second-order moment ($p = 2$):

$$\boxed{\langle (\delta_\ell u)^2 \rangle \sim \left(\frac{\ell}{L}\right)^{2c_1 + 4c_2}.} \tag{18}$$

We now define a normalized increment

$$\boxed{\delta_\ell \nu = \frac{\delta_\ell u}{\sqrt{\langle (\delta_\ell u)^2 \rangle}} = \delta_\ell u \left(\frac{L}{\ell}\right)^{c_1 + 2c_2}.} \tag{19}$$

Replacing $\delta_\ell u = \delta_\ell \nu \, (\ell/L)^{c_1+2c_2}$ in Eq. (20) gives the normalized form of the PDF :

$$\boxed{\begin{aligned}\mathcal{P}_{\delta_\ell \nu}(\delta_\ell \nu) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\alpha}{c_2}+2\alpha^2\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}e^{2\alpha h^\star}}} \\ \times \exp\left[-\alpha\left(\frac{(h^\star-c_1)^2}{2c_2}-h^\star\right)-\frac{1}{2}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}e^{2\alpha h^\star}\right],\end{aligned}} \tag{20}$$

where $h^\star$ is such that $f'(h^\star, \alpha, \delta_\ell u) = 0$.

This normalization ensures that

$$\langle (\delta_\ell \nu)^2 \rangle = 1, \tag{21}$$

for all scales ℓ/L, so that all normalized PDFs can be directly compared across scales.

2.3 Case A: assuming $h^\star = c_1$

From (8),

$$f(h^\star, \alpha, \delta_\ell \nu)\Big|_{h^\star=c_1} = \alpha c_1 + \frac{1}{2}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{4c_2}, \tag{22}$$

and

$$f''(h^\star, \alpha, \delta_\ell \nu)\Big|_{h^\star=c_1} = \frac{\alpha}{c_2} + 2\alpha^2\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{4c_2}. \tag{23}$$

Plug into (10):

$$\boxed{\begin{aligned}\mathcal{P}_{\delta_\ell \nu}(\delta_\ell \nu) \approx \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\sqrt{\frac{2\pi}{\frac{\alpha}{c_2}+2\alpha^2\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{4c_2}}} \\ \times \exp\left[-\alpha c_1-\frac{1}{2}\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{4c_2}\right].\end{aligned}} \tag{24}$$

2.4 Case B: solving $f'(h, \alpha, \delta_\ell \nu) = 0$ via the Lambert W function

From (8) with $\delta_\ell u = \delta_\ell \nu (\ell/L)^{c_1+2c_2}$,

$$\frac{\partial f}{\partial h}(h, \alpha, \delta_\ell \nu) = \alpha\left(\frac{h-c_1}{c_2}-1\right)+\alpha\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}e^{2\alpha h}. \tag{25}$$

Set to zero:

$$h-(c_1+c_2) = -c_2\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}e^{2\alpha h}. \tag{26}$$

Multiply by $-2\alpha e^{-2\alpha h}$:

$$(-2\alpha)(h-c_1-c_2)e^{-2\alpha h} = 2\alpha c_2\left(\frac{\delta_\ell\nu}{\sigma}\right)^2\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}e^{2\alpha(c_1+c_2)}. \tag{27}$$

Using

$$\left(\frac{\ell}{L}\right)^{2(c_1+2c_2)}e^{2\alpha(c_1+c_2)} = \left(\frac{\ell}{L}\right)^{2c_2},$$

let

$$u = -2\alpha(h-c_1-c_2).$$

Then

$$ue^u = 2\alpha c_2 \left(\frac{\delta_\ell \nu}{\sigma}\right)^2 \left(\frac{\ell}{L}\right)^{2c_2}. \tag{28}$$

Hence

$$\boxed{h^\star = (c_1 + c_2) - \frac{1}{2\alpha} W\left(2\alpha c_2 \left(\frac{\delta_\ell \nu}{\sigma}\right)^2 \left(\frac{\ell}{L}\right)^{2c_2}\right)} \tag{29}$$

Moreover,

$$f''(h, \alpha, \delta_\ell \nu) = \frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell \nu}{\sigma}\right)^2 \left(\frac{\ell}{L}\right)^{2(c_1+2c_2)} e^{2\alpha h} = \frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell \nu}{\sigma}\right)^2 \left(\frac{\ell}{L}\right)^{2c_2} e^{2\alpha(h - c_1 - 2c_2)}. \tag{30}$$

Insert $h = h^\star$ into (10):

$$\boxed{\begin{aligned} \mathcal{P}_{\delta_\ell \nu}(\delta_\ell \nu) \approx \frac{1}{2\pi\sigma} \sqrt{\frac{\alpha}{c_2}} \sqrt{\frac{2\pi}{\frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{\delta_\ell \nu}{\sigma}\right)^2 \left(\frac{\ell}{L}\right)^{2c_2} e^{2\alpha h^\star}}} \\ \times \exp\left[-\alpha\left(\frac{(h^\star - c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}\left(\frac{\delta_\ell \nu}{\sigma}\right)^2 \left(\frac{\ell}{L}\right)^{2c_2} e^{2\alpha h^\star}\right]. \end{aligned}} \tag{31}$$

2.5 Visualization of the obtained PDF for different ℓ/L

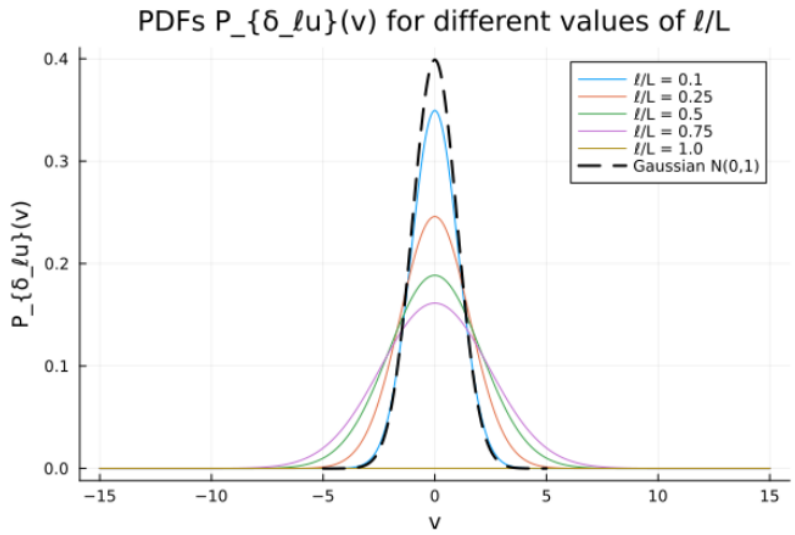


Figure 1: Probability density functions $\mathcal{P}_{\delta_\ell u}(\nu)$ for different values of ℓ/L, compared with a standard Gaussian $N(0, 1)$.

To illustrate the influence of the scale ratio ℓ/L, Figure 1 shows the PDFs $\mathcal{P}_{\delta_\ell u}(\nu)$ as a function of ν for several values of ℓ/L. As ℓ/L decreases, the distributions become more peaked and develop heavier tails.

References

[1] L. Chevillard, B. Castaing, A. Arneodo, E. Lévêque, J.-F. Pinton, and S. G. Roux, *A phenomenological theory of Eulerian and Lagrangian velocity fluctuations in turbulent flows*, C. R. Physique, **13**, 899–928 (2012).